The Journal of Sexual Medicine, 2025, **22**, 645–651
https://doi.org/10.1093/jsxmed/qdaf026
Advance access publication date 25 February 2025
Original Research

Examining gender-specific mental health risks after gender-affirming surgery: a national database study

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Abstract

Background: Transgender individuals face heightened psychological distress, including depression, anxiety, and suicidal ideation, partly due to stigma and lack of gender affirmation.
Aim: To evaluate mental health outcomes in transgender individuals with gender dysphoria who have undergone gender-affirming surgery, stratified by gender and time since surgery.
Methods: This retrospective study utilized the TriNetX database, analyzing U.S. patients aged ≥18 with gender dysphoria (International Classification of Diseases, Tenth Revision [ICD-10] F64) between June 2014 and June 2024. Six cohorts were created based on gender and surgery status: Cohorts A-D included patients with or without surgery, and Cohorts E-F allowed for gender comparison among those with surgery. Propensity score matching controlled for age, race, and ethnicity. Mental health outcomes included depression, anxiety, suicidal ideation, substance use disorder, and body dysmorphic disorder, assessed over two years post-surgery using clinician-verified ICD-10 codes. Body dysmorphic disorder (BDD) was analyzed separately and not conflated with gender dysphoria cohorts to ensure the distinction between these conditions**.** Statistical analysis employed risk ratios, with $P < 0.05$ deemed significant.
Outcomes: Primary outcomes were differences in mental health disorders, specifically depression, anxiety, suicidal ideation, body-dysmorphic disorder, and substance use disorder, among transgender individuals' post-surgery.
Results: From 107 583 patients, matched cohorts demonstrated that those undergoing surgery were at significantly higher risk for depression, anxiety, suicidal ideation, and substance use disorders than those without surgery. Males with surgery showed a higher prevalence of depression (25.4% vs. 11.5%, RR 2.203, $P < 0.0001$) and anxiety (12.8% vs. 2.6%, RR 4.882, $P < 0.0001$). Females exhibited similar trends, with elevated depression (22.9% vs. 14.6%, RR 1.563, $P < 0.0001$) and anxiety (10.5% vs. 7.1%, RR 1.478, $P < 0.0001$). Feminizing individuals demonstrated particularly high risk for depression (RR 1.783, $P = 0.0298$) and substance use disorders (RR 1.284, $P < 0.0001$).
Clinical implications: Findings suggest the necessity for gender-sensitive mental health support following gender-affirming surgery to address post-surgical psychological risks.
Strengths and Limitations: By leveraging ICD-10 codes, we provide a more accurate representation of patient demographics and clinical outcomes, minimizing recall and reporting biases that often limit survey-based research. Limitations include the inability to account for unmeasured confounders such as social support.
Conclusion: Gender-affirming surgery, while beneficial in affirming gender identity, is associated with increased risk of mental health issues, underscoring the need for ongoing, gender-sensitive mental health support for transgender individuals' post-surgery.

Keywords: transgender; gender identity; gender dysphoria; gender-affirming surgery; mental health; TriNetX.

Introduction

Transgender individuals—those who experience a mismatch between their gender identity and the sex assigned at birth—face a heightened risk of psychological distress and related challenges, including suicidal tendencies [1–4]. This increased risk is thought to arise partly from transgender individuals' heightened exposure to stigma-related stress, often referred to as minority stress [5, 6]. Additionally, it may be linked to the stress of not receiving gender affirmation, which involves the accurate acknowledgment and validation of their gender identity [3, 7]. For some transgender individuals, this distress reaches a clinical threshold known as gender dysphoria, defined as significant discomfort or distress stemming from an incongruence between one's experienced or expressed gender and their assigned sex at birth [8, 9]. To relieve the stress associated with the ongoing mismatch between one's gender identity and assigned sex, a growing number of transgender individuals pursue gender-affirming medical treatments, such as hormone therapy and gender-affirming surgeries.

Despite increasing support for gender-affirming medical interventions to alleviate distress in transgender individuals experiencing gender incongruence, the long-term mental health outcomes associated with these interventions remain largely unclear. Much of the available research is based on small sample sizes, cross-sectional designs, and self-reported data on treatment exposure and mental health outcomes,

Received: November 6, 2024. **Revised:** January 25, 2025. **Accepted:** January 28, 2025
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which can introduce biases and limit the reliability of findings [4, 10–13]. A meta-analysis of small-scale studies, primarily cross-sectional, suggested a positive association between self-reported hormone therapy and gender-affirming surgery with improved mental health outcomes [10]. However, these studies are often limited by short follow-up periods and lack of control for confounding variables, making it challenging to establish causative links over time.

Existing studies also fall short due to non-representative sampling and limited longitudinal data, leaving critical questions unanswered about the association between the duration since gender-affirming treatment and mental health outcomes among transgender individuals. Moreover, the absence of probability-based data on the prevalence of mood and anxiety disorders within this population, as compared to the general population, reflects a gap in understanding the true impact of these interventions [14].

Our study seeks to address these gaps by utilizing the TriNetX database and International Classification of Diseases, Tenth Revision (ICD-10) codes, which allow for a more comprehensive, clinician-verified assessment of gender dysphoria and related mental health outcomes across a large, nationally representative cohort. Unlike previous studies that rely on self-reported data and smaller, institution-based samples, our methodology leverages robust, real-world data to enable a more accurate and generalizable understanding of mental health outcomes following gender-affirming surgery. The objectives of this study are threefold: (1) to assess mental health outcomes in transgender individuals with gender dysphoria who have undergone gender-affirming surgery compared to those who have not, (2) to explore gender-specific mental health differences among those who have received gender-affirming surgery, and (3) to evaluate whether mental health outcomes vary based on the length of time since undergoing surgery. By addressing these objectives, this study aims to provide valuable insights into the mental health impacts of gender-affirming surgery, contributing to more informed and supportive care for transgender individuals.

Methods

Data source

This study utilized the TriNetX database, a global health research network managed by a private organization, providing access to de-identified patient data from over 64 U.S.-based healthcare organizations, including a mix of public and private institutions. The database encompasses data from more than 113.4 million patients, aggregated from electronic medical records (EMRs), claims, and other healthcare data sources, ensuring standardized and comprehensive documentation. Organizations contribute their data to support research initiatives, improve healthcare outcomes, and leverage analytics for quality improvement. This study was deemed exempt from Institutional Review Board (IRB) oversight as it exclusively involved de-identified patient data.

Study design and population

The retrospective study selected patients from June 12, 2014, to June 12, 2024 of U.S. patients. To be included, all patients had to be 18 years or older with a diagnosis of gender dysphoria, as identified by the ICD-10 code F64. This criterion was chosen based on literature highlighting elevated mental health concerns for transgender and nonbinary patients with gender dysphoria [15, 16]. Gender-affirming surgery cohorts consisted of patients with a documented diagnosis of gender dysphoria who had undergone specific gender-affirming surgical procedures. For transmen, this primarily included mastectomy (chest masculinization surgery, CPT codes 19 303 and 19 304), while for transwomen, this encompassed a range of feminizing procedures such as tracheal shave (CPT code 31899), breast augmentation (CPT code 19325), and vaginoplasty (CPT codes 57 335 and 55 970). These surgeries were identified using clinician-verified CPT codes within the TriNetX database, allowing for precise classification.

Classification of cohorts

We classified patients using the gender documented in the EMRs within the TriNetX database, recognizing that this documentation may reflect either natal sex or gender identity, depending on how it was recorded. To minimize potential misclassification, we identified transgender individuals using the ICD-10 code F64 (gender dysphoria) and categorized them into six cohorts.

- **Cohort A**: Patients documented as male (which may indicate natal sex or affirmed gender identity), aged ≥18 years, with a prior diagnosis of gender dysphoria, who had undergone gender-affirming surgery.
- **Cohort B**: Male patients with the same diagnosis but without surgery.
- **Cohort C**: Patients documented as female, aged ≥18 years, with a prior diagnosis of gender dysphoria, who had undergone gender-affirming surgery.
- **Cohort D**: Female patients with the same diagnosis but without surgery.
- **Cohort E**: Transgender male patients who underwent masculinizing gender-affirming regardless of a previous documented diagnosis of gender dysphoria
- **Cohort F**: Transgender female patients who underwent feminizing gender-affirming surgery regardless of a previous documented diagnosis of gender dysphoria.

Cohorts E and F include transgender patients who underwent gender-affirming surgery but lacked a documented diagnosis of gender dysphoria, unlike Cohorts A and C, which specifically require this diagnosis for inclusion. This distinction allows for the evaluation of mental health outcomes in a broader transgender population, encompassing individuals who sought surgery without meeting the formal diagnostic criteria for gender dysphoria. By comparing these cohorts, the study provides unique insights into how mental health outcomes may differ based on diagnostic status. While longitudinal data at the individual level were unavailable, mental health outcomes were assessed in a cross-sectional manner using diagnoses recorded before and after surgery within the database. Risk for mental health outcomes was assessed for all cohorts over two years following surgery, based on findings from the 2015 US Transgender Survey that highlighted significant adverse mental health outcomes occurring within this timeframe [17]. Mental health outcomes were determined using validated tools administrated by the doctors and healthcare organizations, with the results recorded using corresponding ICD-10 codes.

Mental health outcome assessment

Mental health outcomes in this study were assessed using clinician-verified International Classification of Diseases, Tenth Revision (ICD-10) diagnostic codes, as recorded in the EMRs within the TriNetX database. These diagnoses were established by healthcare professionals during clinical encounters and documented in the EMRs of participating healthcare organizations. This approach eliminates the reliance on self-report measures, ensuring that diagnoses such as depression, anxiety, suicidal ideation, substance use disorder, and body dysmorphic disorder are based on clinical evaluations rather than patient-reported symptoms or survey items. By utilizing ICD-10 codes, we sought to enhance the validity and reliability of the data, addressing the limitations of bias and subjectivity inherent in self-reported mental health measures.

Study outcomes

The outcomes for our study were chosen based on studies that highlighted mental health outcomes that exist within the transgender patient population [18–20]. Patients included in the analysis had no documented mental health disorder diagnoses prior to the index date. The absence of longitudinal patient trajectories in the database limited within-patient tracking, and outcomes were evaluated cross-sectionally based on ICD-10 diagnoses documented at two time points: pre- and post-surgery. The database does not include explicit information on sex assigned at birth, relying instead on documented demographic data as "male" or "female."

Propensity score matching

To control for potentially confounding factors, propensity score matching was utilized. In our study, we propensity matched for age, race, and ethnicity, criteria identified in the literature as risk factors for mental health in this population [21, 22].

Statistical analysis

Data analysis was conducted using the TriNetX software platform, which facilitates statistical computations and cohort comparisons. Risk ratios (RRs) with 95% confidence intervals (CIs) were calculated to assess the relative risk of mental health outcomes between cohorts. Statistical significance was determined with a threshold of $P < 0.05$. Additional tables summarizing demographic and outcome data were generated using Microsoft Excel to provide a comprehensive overview of the results.

Results

Our team identified 107 583 patients aged ≥18 with a previous diagnosis of gender dysphoria using the TriNetX Database United States Collaborative Network. Initially, Cohort A included 2774 male patients with gender dysphoria and gender-affirming surgery; Cohort B included 48 090 male patients with gender dysphoria but no gender-affirming surgery; Cohort C included 3358 female patients with gender dysphoria and gender-affirming surgery; Cohort D included 67 579 female patients with gender dysphoria but no gender-affirming surgery; Cohort E included 3790 transgender male patients who underwent gender-affirming surgery but did not have a documented diagnosis of gender dysphoria; Cohort F included 4643 transgender female patients who underwent gender-affirming surgery but did not have a documented diagnosis of gender dysphoria. The demographics for each cohort before and after propensity score matching is attached to the supplementary tables.

After propensity score matching of cohorts A and B, each cohort had 2774 patients of similar race, ethnicity, and age at index (**Supplementary document:** Table S1). Compared to male patients with a diagnosis of gender dysphoria only, those with gender affirmation surgery were at significantly higher risk for depression, anxiety, suicidal ideation, and substance use disorders. However, neither cohort was at increased risk for body dysmorphic disorder (Table 1). Male patients with gender-affirming surgery had a 25.4% rate of depression, compared to 11.5% for those without surgery (RR 2.203, 95% CI 1.477-3.287, $P < 0.0001$). Male patients with surgery had 4.882 times the risk of anxiety (12.783% vs. 2.618%, RR 4.882, 95% CI 4.505-5.29, $P < 0.0001$) compared to those who did not receive surgery (12.783% vs. 2.618%, RR 4.882, 95% CI 4.505-5.29, $P < 0.0001$). Both groups had the same risk for body dysmorphic disorder (0.4% vs. 0.4%, RR 1.001, 95% CI 0.417-2.402, $P = 0.9974$).

After propensity score matching of Cohorts C and D, each cohort had 3358 female patients of similar age at index, race, and ethnicity (**Supplementary document:** Table S2). Female patients with gender dysphoria and a history of gender-affirming surgery had significantly higher risks for depression, anxiety, suicidal ideation, and substance use disorders compared to those with a diagnosis of gender dysphoria only. However, neither group was at an increased risk for body dysmorphic disorder (Table 2). Females with gender-affirming surgery had a 22.9% rate of depression, compared to 14.6% for those without surgery (RR 1.563, 95% CI 1.422-1.717, $P < 0.0001$). Compared to those without surgery, females who had undergone gender-affirming surgery had a 1.478 times higher risk of anxiety (10.496% vs. 7.098%, RR 1.478, 95% CI 1.214-1.797, $P < 0.0001$), a 2.357 times higher risk of suicidal ideation (19.811% vs. 8.402%, RR 2.357, 95% CI 1.579-3.515), and a 2.712 times higher risk of substance use disorder (19.322% vs. 7.123%, RR 2.712, 95% CI 1.439-3.217). Both groups had the same risk for body dysmorphic disorder (0.3%) (Table 2).

To assess gender disparities in mental health outcomes in transgender patients who underwent gender-affirming surgery but lacked a documented diagnosis of gender dysphoria, we compared Cohorts E and F. After propensity score matching, both cohorts included 3607 patients who were similar at index, in age, race, and ethnicity (**Supplementary document:** Table S3). Transgender men who had undergone gender-affirming surgery were at higher risk of most mental health issues compared to transgender women. Specifically, transgender men had a 1.58 times higher risk of anxiety (14.1% vs. 8. 9%, RR 1.580, 95% CI 0.845-2.134, $P = 0.0002$), a 1.186 times higher risk of suicidal ideation (5.5% vs. 4.6%, RR 1.186, 95% CI 0.97-1.449, $P = 0.0358$), and a 1.284 times higher risk of substance use disorder (14.4% vs. 11.2%, RR 1.284, 95% CI 1.137-1.45, $P < 0.0001$). Among the five outcomes, the relative risk was highest for depression among transgender men compared to transgender women (RR 1.783, 95% CI 1.327-2.389, $P = 0.0298$). Both cohorts were at the same risk for body dysmorphic disorder (Table 3).

Table 1. Outcomes for male patients with a previous diagnosis of gender dysphoria following gender-affirming surgery (cohort A) vs. male patients with gender dysphoria only (cohort B) after propensity score matching.

Outcomes	Cohort A	Cohort B	RR (95% CI)	p-value
Depression	25.4%	11.5%	2.203 (1.477, 3.287)	<0.0001
Anxiety	12.8%	2.6%	4.882 (4.505, 5.29)	<0.0001
Suicidal ideation	3.4%	2.5%	1.356 (0.984, 1.868)	0.0002
Substance use disorder	19.0%	8.2%	2.299 (2.158, 2.45)	<0.0001
Body dysmorphic disorder	0.4%	0.4%	1.00 (0.417, 2.402)	0.9974

RR: relative risk; CI: confidence interval.

Table 2. Female patient outcomes following gender-affirming surgery with a previous diagnosis of gender dysphoria (cohort C) vs. female patient outcomes with a previous diagnosis of gender dysphoria only (cohort D) after propensity score matching.

Outcomes	Cohort C	Cohort D	RR (95% CI)	p-value
Depression	22.9%	14.6%	1.563 (1.422, 1.717)	<0.0001
Anxiety	10.5%	7.1%	1.478 (1.214, 1.797)	<0.0001
Suicidal ideation	19.8%	8.4%	2.357 (1.579, 3.515)	0.0401
Substance use disorder	19.3%	7.1%	2.712 (1.439, 3.217)	0.0193
Body dysmorphic disorder	0.3%	0.3%	1.00 (0.416, 2.406)	0.9995

RR: relative risk; CI: confidence interval.

Table 3. Outcomes of transgender males without documented gender dysphoria following gender-affirming surgery (cohort E) vs. transgender females without documented gender dysphoria following gender-affirming surgery (cohort F) after propensity score matching.

Outcomes	Cohort E	Cohort F	RR (95% CI)	p-value
Depression	44.2%	24.7%	1.789 (1.327, 2.389)	0.0298
Anxiety	14.1%	8.9%	1.580 (0.845, 2.134)	0.0002
Suicidal ideation	5.5%	4.6%	1.186 (0.97, 1.449)	0.0358
Substance use disorder	14.4%	11.2%	1.284 (1.137, 1.45)	<0.0001
Body dysmorphic disorder	0.3%	0.3%	1.00 (0.416, 2.405)	1.000

RR: relative risk; CI: confidence interval.

Discussion

The findings of this study underscore a pressing need for enhanced mental health guidelines tailored to the needs of transgender individuals following gender-affirming surgery. Our analysis reveals a significantly elevated risk of mental health disorders—including depression, anxiety, suicidal ideation, and substance use disorder—post-surgery among individuals with a prior diagnosis of gender dysphoria. Importantly, however, our results indicate no increased risk of body dysmorphic disorder following surgery, suggesting that these individuals generally experience satisfaction with their body image and surgical outcomes. Notably, the heightened risk of mental health issues post-surgery was particularly pronounced among individuals undergoing feminizing transition compared to masculinizing transition, emphasizing the necessity for gender-sensitive approaches even after gender-affirming procedures.

By excluding patients with documented pre-existing mental health diagnoses, this study sought to ensure that identified mental health outcomes likely represented new or emergent conditions rather than pre-existing disorders. This methodological approach was critical to focusing on the relationship between gender-affirming surgery and mental health. However, we acknowledge that this approach, relying solely on ICD-10 codes, may not fully account for undiagnosed or subclinical conditions prior to surgery. These emergent mental health issues may result from a multifactorial interplay of social, psychological, and physiological factors, including social support systems, environmental stressors, hormonal changes, surgical outcomes, and the broader psychosocial adjustments involved in transitioning.

Comparison with previous studies

When evaluating these findings within the context of previous research, it is crucial to recognize the limitations inherent in studies that rely primarily on survey data, such as those analyzed by Marano et al. and Almazan and Keuroghlian [13, 23]. These studies, using data from the U.S. Transgender Survey, underscore the psychosocial benefits of gender-affirming surgeries, including reductions in depression, anxiety, and suicidal ideation, while emphasizing the importance of aligning physical appearance with gender identity to improve mental health. However, survey-based studies are limited by self-reported data, which may introduce response bias and lack clinical validation, potentially limiting the generalizability of their findings [24]. Our study diverges by using a national database of de-identified clinical data, enabling a more comprehensive and representative examination of real-world mental health outcomes across diverse demographics. This approach allows us to capture more nuanced insights into mental health risks, particularly the heightened susceptibility to depression, anxiety, suicidal ideation, and substance use disorder in transwomen individuals' post-surgery. This divergence from survey-based findings highlights the need for gender-sensitive mental health strategies that extend beyond the surgical intervention itself.

Implications for mental health care

Despite the observed increase in mental health issues, gender-affirming surgery remains essential in aligning transgender individuals' physical appearance with their gender identity, offering significant psychological benefits [8, 19]. Research,

such as that conducted by Park et al., has documented long-term satisfaction and mental health improvements in patients who have undergone gender-affirming surgeries over decades [25]. These enduring benefits underscore the necessity for mental health practitioners to recognize and address these specific challenges, ensuring that post-surgical mental health care is both accessible and gender-responsive.

It is also crucial to acknowledge that transgender individuals seek mental health support for a wide range of issues, not solely those related to gender identity. The lifelong impact of minority stress continues to affect transgender individuals' experiences of depression and anxiety even after transitioning [3, 26, 27]. Barriers to mental health care, including discrimination within healthcare settings, exacerbate these mental health challenges, fostering systemic distrust and reducing access to necessary services [28–30]. Our findings highlight that anxiety is particularly prevalent among transgender men post-surgery, while substance use disorder is more common among transgender women, reflecting gender-specific mental health risks. For transgender women, societal pressures to conform to traditional female roles and the pervasive devaluation of femininity may contribute to heightened stress, emotional distress, and, ultimately, increased reliance on substance use as a coping mechanism [31, 32]. Conversely, transgender men may encounter societal expectations to suppress emotions, aligning with traditional masculine norms, which can heighten anxiety as they navigate their new gender identity.

Hierarchical criteria and mental health diagnoses

An important consideration in interpreting our findings is the hierarchical nature of psychiatric diagnoses, as specified in the DSM. This framework often precludes standalone diagnoses of anxiety or depression if these symptoms are deemed to be better explained by another superior diagnosis, such as gender dysphoria [33]. Consequently, symptoms of anxiety or depression that co-occur with gender dysphoria may be subsumed under the latter diagnosis, particularly in pre-surgical contexts. Following gender-affirming surgery, the alleviation of distress related to gender incongruence may enable the reclassification of these symptoms as independent diagnoses. This diagnostic shift could contribute to the observed increase in mental health diagnoses post-surgery, not as a reflection of adverse surgical outcomes but rather as a reconceptualization of symptoms within the care pathway. Including this perspective enhances our understanding of the study's findings and emphasizes the need for nuanced mental health assessments tailored to the unique trajectories of transgender individuals. Future research should explore how changes in diagnostic frameworks and psychiatric practices influence mental health outcomes in this population.

Future directions

Further research should investigate the complex factors contributing to mental health disparities post-surgery, including social support, family acceptance, societal stigma, and pre-existing mental health conditions. Prospective, longitudinal studies are needed to track changes in mental health from pre-surgery through long-term follow-up, providing greater clarity on the causal impact of gender-affirming surgery. Additionally, examining how systemic factors, such as healthcare policy, insurance coverage, and provider training, influence access to care would offer critical insights into improving equity and effectiveness in mental health care for transgender individuals.

While the limitations of this study cannot be fully overcome with the current data available from the TriNetX database, it is important to carefully interpret the conclusions within the context of these constraints. The retrospective design and reliance on de-identified, aggregated data restrict our ability to establish causation or continuously follow individuals across healthcare systems. Furthermore, potential misclassification of mental health outcomes due to undocumented pre-existing conditions or incomplete follow-up outside the TriNetX network remains a limitation. Despite these challenges, our analytic design offers valuable insights into associations between gender-affirming surgery and mental health outcomes at a population level, leveraging clinician-verified ICD-10 codes to enhance diagnostic reliability compared to self-reported data. However, this approach cannot capture nuanced individual trajectories or address disparities in access to care that may influence the likelihood of receiving a diagnosis. Future research should employ longitudinal designs with continuous follow-up to better address these limitations, allowing for more robust evaluations of the relationship between gender-affirming care and mental health outcomes.

Study limitations

While this study offers critical insights into the mental health challenges experienced by transgender individuals following gender-affirming surgery, several limitations must be acknowledged. The TriNetX database, comprising de-identified patient records, restricts patient-level linkage for multiple diagnoses or tracking individual health trajectories, which limits our ability to perform true longitudinal or within-person analyses. Instead, our analysis relied on cross-sectional comparisons of mental health outcomes before and after surgery. While TriNetX aggregates patient data from multiple healthcare organizations within its network, this does not extend to patients who leave the network entirely, potentially leading to incomplete follow-up data. Additionally, data stored in unstructured formats, such as clinical notes, are not included, which may contribute to selection bias. A significant limitation is the potential selection bias inherent in the study population. Individuals pursuing gender-affirming surgery may represent a subgroup experiencing higher levels of psychological distress compared to those who do not seek surgery. This increased baseline distress could inherently elevate the risk of adverse mental health outcomes, independent of the surgical intervention itself. Future research should consider methods to account for these pre-existing differences to better understand the true impact of surgery on mental health outcomes.

One significant limitation is the binary classification of gender within the TriNetX database, which only records patients as "male" or "female" in its demographic data. This excludes non-binary individuals and others who do not align with binary gender categories, limiting the inclusivity and representativeness of the study. Furthermore, the database does not include explicit information on sex assigned at birth, legal gender changes, or affirmed gender identities, which prevents more nuanced subgroup analyses. This limitation underscores the importance of developing future data systems that allow for broader gender identity categories to support more inclusive research.

While our use of clinician-verified ICD-10 codes ensures objective and standardized diagnoses, these codes are reliant

on the clinical expertise and practices of healthcare professionals, which may vary across organizations. This variability introduces potential inconsistencies in diagnosis accuracy. Moreover, differences in healthcare access likely influence the likelihood of receiving formal mental health diagnoses. Patients undergoing surgery often have greater access to healthcare services, including mental health care, compared to those who do not. This may lead to higher rates of mental health diagnoses in surgical cohorts, independent of actual differences in mental health status, introducing potential surveillance bias.

Another limitation is the potential misclassification of individuals in the "no-surgery" cohort. The TriNetX database captures surgical history only from participating organizations, which means that patients who underwent gender-affirming surgeries outside of these institutions may have been incorrectly categorized. This limitation may affect the accuracy of our comparisons between surgical and non-surgical cohorts. Future studies with access to centralized and comprehensive data sources are needed to improve the classification of surgical histories.

The criteria for identifying transgender individuals in this study were based on documented diagnoses of gender dysphoria (ICD-10 code F64). This approach excludes transgender or gender-diverse individuals who do not seek medical treatment for gender incongruence, limiting the generalizability of our findings. Additionally, the absence of a comparison group for individuals who sought but had not yet received gender-affirming treatments restricts the study's ability to assess the impact of treatment timing on mental health outcomes. Longitudinal studies that track outcomes before and after treatment are needed to address this gap.

Lastly, mental health treatment utilization serves as an imperfect proxy for mental health itself. Transgender individuals receiving treatment for gender dysphoria are frequently in healthcare settings, where they may encounter more mental health treatment opportunities, potentially skewing utilization rates. Finally, while our findings support the need for accessible gender-affirming treatments, the generalizability of these results may be further limited by systemic factors, including healthcare policies, insurance coverage, and regional differences in provider training. These barriers, which vary significantly across different healthcare systems, can affect transgender individuals' access to adequate mental health care and underscore the necessity for policies that ensure consistent, affirming care.

Conclusion

Our study reveals that both male and female patients with gender dysphoria who undergo gender-affirming surgery are at significantly higher risk for adverse mental health outcomes, including depression, anxiety, suicidal ideation, and substance use disorder, compared to those who do not undergo gender-affirming surgery. This trend persists even after controlling for confounding factors through propensity score matching. Notably, transgender men showed a greater relative risk for these mental health issues compared to transgender women following gender-affirming surgery. Despite the benefits of surgery in alleviating gender dysphoria, our findings underscore the necessity for ongoing mental health support for transgender individuals during their post-surgery trajectories. These results also highlight the critical need for gender-specific care tailored to the unique experiences of male and female populations, respectively, addressing both pre- and post-surgical mental health care to improve overall well-being and prevent any mental illness or diseases.

Acknowledgements

The authors thank Norma A Pérez Raifaisen, MD, DrPH, CPC-ELI-MP, Director of the UTMB Center of Excellence for Professional Advancement and Re-search (COEPAR), for assistance with manuscript preparation.

Author contributions

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Supplementary material

Supplementary material is available at *The Journal of Sexual Medicine* online.

Conflicts of interest

The authors that there are no conflicts of interest regarding the publication of this paper.

Funding

This work was supported in part by the Department of Health and Human Services, Health Resources and Services Administration [grant number 1 D34HP49234-01-00].

Ethical statement

The study was deemed IRB exempt from University of Texas Medical Branch Institutional Review Board.

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The Journal of Sexual Medicine, 2025, 22, 645–651
https://doi.org/10.1093/jsxmed/qdaf026
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